RECONSTRUCT.AI

Converting Scattered Project Activity into Verifiable Handover Intelligence

Project purpose

Reconstruct.ai is a proposed handover-intelligence platform that gathers authorized project activity from code repositories, team discussions, issue trackers and project documents. It organizes that evidence into a time-aware, source-linked view of project decisions, unresolved work, ownership risks and next steps—so a successor can continue work with less guesswork.

Team	Reconstruct.ai
Institution	JSPM’s Rajarshi Shahu College of Engineering (RSCOE), Pune
Team members	Khushi Patil • Dhruv Patil • Yuvraj Kadam • Nimisha Joshi
Repository	github.com/Dhruvplaz-web/reconstruct-ai
Submission stage	Round 1 — Plan and proposal

Official Submission Endorsement: *This consolidated proposal is prepared and submitted by Team Reconstruct.ai (JSPM RSCOE, Pune) for Round 1 of the We Code Coders Three-Round Challenge. All proposed workflows, data models, and milestone commitments have been aligned directly with the hackathon's required core workflow and evaluation criteria.*

1. Understanding of the Problem

Project teams accumulate knowledge across many places: source-code commits, pull requests, issue trackers, chat discussions, design notes and individual working practices. A project handover becomes difficult when a key contributor leaves, graduates, changes responsibilities or becomes unavailable. The next person may inherit the code and task list but not the reasoning behind decisions, hidden dependencies, debugging steps, operational warnings or the history of unresolved issues.

The core problem is therefore not simply a lack of documentation. It is the fragmentation of project context and the difficulty of identifying which information is current, reliable, relevant and safe to share. Traditional handover notes are often manually prepared and can become stale; search tools can retrieve information but may not connect related events or clearly expose conflicting evidence.

Reconstruct.ai addresses this gap by converting authorized project traces into a structured, evidence-linked handover workspace. It is designed to assist human handover—not to replace contributor judgement or automatically declare every extracted statement true.

Problem dimensions

- **Scattered context:** important decisions and blockers are distributed across tools and conversations.
- **Knowledge concentration:** critical files or workflows may depend heavily on one contributor.
- **Unclear project state:** old discussions, current code and open tickets may not agree.
- **Successor uncertainty:** incoming contributors may not know what to inspect or whom to ask.
- **Trust and privacy:** handover material can contain confidential information that should not be visible to every role.

Scope boundary

The initial product focuses on software and project-team handovers, with GitHub/Git history and Slack-style exported discussions as the first practical sources. Jira/Linear and document connectors are planned extensions or optional inputs depending on time and access. The same approach may later support research labs, student organizations and open-source communities, but these are expansion contexts rather than separate MVP builds.

2. Affected Users

USER GROUP	NEED / PAIN POINT	VALUE FROM RECONSTRUCT.AI
Departing contributor	Needs a focused way to transfer critical context without writing a long manual handover.	Targeted questions based on detected ownership gaps and unresolved work; human review before sign-off.
Incoming successor	Needs to understand current architecture, decisions, risks and next actions.	Source-linked briefing, timeline, unresolved items and Day 1–30 task list.
Team lead / project manager	Needs visibility into handover completeness, dependencies and knowledge concentration.	Risk indicators, review status and a structured view of project continuity.
Security / repository owner	Needs to control which records and sensitive fields can be accessed.	Permission-aware ingestion and server-side access filtering, subject to policy configuration.

3. Proposed Solution

Reconstruct.ai is a multi-source handover reconstruction system. It collects project evidence that the team is authorized to use, normalizes events into a common structure, links related decisions and artifacts over time, highlights ownership concentration and unresolved questions, and presents the result as an interactive continuation workspace.

The platform is designed around five principles:

- **Evidence before assertion:** every generated handover claim should link back to one or more source records.
- **Time-aware context:** the system should show when information was created or updated and avoid treating old notes as current without review.
- **Uncertainty is visible:** missing support and possible contradictions should be flagged for human inspection.
- **Human confirmation:** departing contributors and project leads can correct or approve important handover details.
- **Least-privilege access:** sensitive content is filtered on the server according to the viewer's permissions.

Differentiation

The proposed distinction is the combination of repository ownership analysis, cross-source timeline reconstruction, explicit evidence citations, uncertainty/contradiction review and successor-oriented tasks in one workflow. Existing documentation, search and code-guide tools can each address parts of this problem; Reconstruct.ai aims to connect those parts into a handover process. This is a product-positioning statement, not a claim that other products lack all of these capabilities.

4. Complete Product Workflow



Figure 1: Complete 5-Step Operational Handover Workflow

1 — Connect: A manager selects the project and departing contributor, authorizes the data sources and chooses either connected sources or a prepared synthetic demo dataset.

2 — Reconstruct: The ingestion service parses commits, pull requests, issue records and permitted discussion exports. It extracts candidate decisions, dependencies, blockers and ownership patterns. Git blame concentration is treated as a risk signal, not proof that a person is the only one who understands a module.

3 — Verify: The system links each candidate statement to its supporting source and timestamp. Where two statements appear inconsistent, an NLI model may flag a possible conflict; the interface presents the original evidence for review rather than deciding factual truth automatically.

4 — Review: The departing contributor or project lead reviews high-risk gaps and answers a small set of targeted questions. Corrections and approvals are recorded with their author and time.

5 — Hand over: The platform applies role-based/attribute-based access rules and generates a continuation workspace with a project summary, evidence timeline, unresolved tasks and suggested first-month actions.

5. Main Planned Features

FEATURE	PLANNED CAPABILITY	USER BENEFIT
Multi-source ingestion	Import repository history and authorized discussion/ticket exports; provide a synthetic demo path for evaluation.	Reduces manual collection and avoids requiring judges to connect personal accounts.
Ownership concentration scan	Analyze Git blame / commit history and flag files with high single-author contribution for review.	Helps teams identify modules that may need knowledge transfer.
Temporal evidence graph	Represent people, files, decisions, issues and events with timestamped, traceable relationships.	Shows how project context evolved and where a claim came from.
Evidence-linked handover brief	Generate structured summaries with source references and timestamps.	Lets successors inspect the original evidence instead of relying only on generated text.
Uncertainty / conflict review	Use NLI as a signal to identify potentially inconsistent statements; route uncertain items to review.	Makes disagreement visible and reduces silent merging of conflicting claims.
Targeted leaver questions	Generate a short set of questions from detected gaps, ownership concentration and unresolved items.	Focuses contributor effort on the most important missing context.
Continuation workspace	Show a project overview, timeline/graph, citations, unresolved work and Day 1–30 tasks.	Turns handover information into actionable next steps.
Permission-aware view	Apply server-side access policies and redact restricted fields before returning data.	Supports safer sharing across roles; requires security testing and policy review.



Figure 2: Reconstruct.ai Production Web Interface: 3-Column Continuation Hub, Role Switcher & Grounded Citations

6. Proposed Technology Stack

LAYER	PROPOSED TECHNOLOGY	WHY IT FITS / IMPLEMENTATION NOTE
Frontend	Next.js 14, TypeScript, Tailwind CSS, Radix UI	Responsive dashboard, accessible dialogs/drawers and reusable UI. Version should be aligned with the actual project setup before implementation.
Backend API	Python, FastAPI, Pydantic, Uvicorn	Python ecosystem supports repository parsing, NLP and graph processing; typed request/response schemas simplify API integration.
Ingestion	GitPython; GitHub API/PyGithub; JSON/Markdown parsers	Start with repository cloning/history and local exported files. Live OAuth connectors are optional extensions, not a Round 2 dependency.
Graph / storage	NetworkX; SQLite for MVP; PostgreSQL for later deployment	NetworkX is sufficient for a prototype graph. Start with a simple relational schema; add vector search only if required by tested workflows.
AI / verification	LLM-based structured extraction; Hugging Face DeBERTa-v3 NLI	LLM output must be schema-validated and citation-grounded. NLI is a possible-conflict classifier, not a truth detector. Model/provider choice depends on availability and evaluation.
Security / testing	Server-side ABAC policy checks; PyTest; frontend tests	Enforce least privilege before serialization; test access denial, redaction, secret handling and audit logging.

7. Team Responsibilities

The responsibilities below consolidate the four individual contributor dossiers. Each member owns a primary area while integration and review remain shared responsibilities.

MEMBER / ROLE	PRIMARY RESPONSIBILITIES	ROUND DELIVERABLES
Khushi Patil **Team Lead & Head of Research / Product Strategy**	Problem validation, user taxonomy, research framing, competitor review, sprint planning, acceptance criteria, team coordination and submission compliance.	R1: problem dossier, scope, evaluation plan and role coordination. R2: UAT scenarios, integration coordination and user feedback. R3: final walkthrough, product review and accessibility/security coordination.
Dhruv Patil **AI Development & Core Engineering Lead**	Structured extraction design, prompt schemas, decision/dependency extraction, grounding and NLI conflict-signalling pipeline, synthetic benchmark design.	R1: AI architecture, prompt/schema plan and benchmark specification. R2: extraction, citation linker and NLI prototype. R3: threshold evaluation, UI integration and hallucination guardrails.
Yuvraj Kadam **Lead Full-Stack Frontend & UI/UX Engineer**	Next.js application, responsive layouts, continuation workspace, onboarding Kanban, citation sidecar, leaver review interface and accessibility.	R1: wireframes, component plan and design tokens. R2: functional web client, demo interface and API integration. R3: graph visualization, responsive polish and keyboard accessibility.
Nimisha Joshi **Systems Backend, Data Ingestion & Security Architect**	FastAPI services, repository parsing, Git ownership analysis, multi-source ingestion schemas, provenance records and server-side ABAC/redaction.	R1: schemas, provenance model, ownership logic and policy design. R2: backend, repo ingestion, discussion export parser and demo loader. R3: automated tests, security review and approval-verification enhancement if feasible.

Shared responsibilities

- All members participate in code review, integration testing, documentation and judge preparation.
- The team lead coordinates scope and milestones; technical leads remain accountable for their owned subsystems.
- The team will not describe planned features as completed until a working implementation and supporting evidence exist.

8. Three-Round Development Plan

ROUND	FOCUS	PLANNED DELIVERABLES	EXIT CHECK
Round 1 — Plan	Validate the problem, define MVP scope and design the architecture.	Consolidated proposal; workflow and architecture; initial data schema; wireframes; synthetic test scenario; team responsibilities.	All required proposal sections reviewed; scope and assumptions approved by team.
Round 2 — Build	Deliver a working end-to-end MVP using a controlled dataset.	Repo ingestion; ownership scan; structured extraction; evidence citations; basic timeline/workspace; targeted review questions; demo dataset.	A judge can load demo data and complete one handover flow end-to-end.
Round 3 — Finish	Improve reliability, usability, privacy and explainability.	Conflict/uncertainty indicators; role-aware access controls; tests; responsive/accessibility improvements; walkthrough video and limitation notes.	Demonstration is repeatable; core flows and access rules pass documented tests.

9. Expected Impact and Evaluation Metrics

The project's impact will be evaluated through a small pilot rather than presented as a guaranteed percentage improvement. The team will compare a baseline handover process with a Reconstruct.ai-assisted handover using the same sample project and successor tasks.

METRIC	HOW TO MEASURE	SUCCESS SIGNAL
Time to first meaningful task	Measure elapsed time for a new contributor to identify and begin a predefined task using the baseline and the generated handover.	Lower time with no loss in task correctness.
Handover completeness	Use a checklist of required decisions, dependencies, open issues, ownership and operational notes; count items supported by evidence.	More checklist items are discoverable and source-linked.
Ownership-risk coverage	Compare manually identified critical files with files flagged by the ownership scan; inspect false positives and false negatives.	Useful risk flags with explainable ownership evidence.
Evidence traceability	Audit a sample of generated claims and verify that each opens the correct source record and timestamp.	Claims are reviewable and unsupported claims are clearly marked.
Review effort	Record time spent by the departing contributor reviewing and answering system-generated questions.	Less effort than preparing a full manual handover, without overstating a fixed time target.
Privacy / access control	Run role-based test cases for allowed and denied records; inspect logs and API responses for restricted fields.	No unauthorized disclosure in the defined test suite.

Impact statement

Expected impact: reduce avoidable context-search effort, make critical project knowledge easier to locate, surface ownership and continuity risks earlier, and provide successors with an actionable, evidence-linked starting point. The pilot will establish measured baselines before the team reports numerical improvement.

10. Future Scope

- **Additional integrations:** GitLab, Jira/Linear, Discord and approved document repositories after the core GitHub/export workflow is stable.
- **Continuous decision capture:** optional bots or integrations that capture approved decisions as they occur, with explicit workspace consent and retention controls.
- **Enterprise identity and policy:** SSO, organization-managed roles, audit export and configurable retention policies.
- **Private deployment:** self-hosted inference and storage options for organizations with strict data residency requirements.
- **Research and education workflows:** adapt the evidence model for laboratory protocols, datasets, student clubs and open-source maintainer transitions.

11. Complete Team and Participant Details

NAME	ROLE	INSTITUTION	EMAIL AS REGISTERED
Khushi Patil	Team Lead & Head of Research / Product Strategy	JSPM RSCOE, Pune	khushi@reconstruct.ai
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Yuvraj Kadam	Lead Full-Stack Frontend & UI/UX Engineer	JSPM RSCOE, Pune	yuvraj@reconstruct.ai
Nimisha Joshi	Systems Backend, Data Ingestion & Security Architect	JSPM RSCOE, Pune	nimisha@reconstruct.ai

Project repository: https://github.com/Dhruvplaz-web/reconstruct-ai

Submission Verification: *All participant identities, institution affiliations (JSPM RSCOE, Pune), and primary roles have been verified and confirmed across all project records. The official code repository is publicly accessible to the judges, and development will proceed strictly during the permitted hackathon build window.*

Implementation Assumptions and Responsible-Use Notes

To keep the proposal technically defensible, the following constraints apply to the design and evaluation:

- **Git blame concentration** (for example, a high single-author share) is a heuristic risk indicator. It does not prove that the contributor is the only person with relevant knowledge or that the file is operationally critical.
- **A temporal relationship** between a chat message and a commit does not by itself prove causality. The graph should distinguish observed links from inferred links and show the evidence behind each inference.
- **Natural Language Inference** can classify pairs of text as potentially entailing, contradicting or being unrelated. It cannot establish which statement is factually correct; timestamps, source authority and human review remain necessary.
- **A provenance model** such as W3C PROV-DM records entities, activities, agents and derivation relationships. It does not, by itself, make records immutable or cryptographically tamper-proof.
- **Access control** must be enforced on the backend before data is serialized or returned. UI masking alone is not a security boundary. The implementation requires negative tests for restricted access and secret leakage.
- **All AI-generated claims** should include source references, be schema-validated and be marked uncertain when evidence is missing. The system must not invent missing handover details.
- **The synthetic demo dataset** must contain fictional project data and secrets, not real employee or organizational information.

Selected technical references

- **W3C, PROV-DM: The PROV Data Model:** https://www.w3.org/TR/prov-dm/
- **Nonaka, I. & Takeuchi, H., The Knowledge-Creating Company:** Foundational reference for the SECI knowledge-creation model.
- **Hugging Face model documentation for DeBERTa / NLI:** Model selection and evaluation should be confirmed against the chosen checkpoint and task.
- **IEEE Software / ACM Digital Library:** Empirical studies on turnover vulnerabilities, knowledge concentration, and documentation maintenance in agile teams.

End of Round 1 proposal • Team Reconstruct.ai • We Code Coders Three-Round Challenge